Susana Loureiro

ML Engineer

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Summary

ML Engineer building end-to-end machine learning systems, from fine-tuned LLMs to real-time computer vision pipelines. BSc in Informatics Engineering at ISEP, specialising in deep learning, NLP, and computer vision. AWS Certified in Machine Learning Engineer Associate, Solutions Architect Associate, and AI Practitioner. Strong software engineering foundation enabling end-to-end development from model training to production deployment.

Experience

ML Engineer Intern — February 2026 – Present
DevScope — Porto, Portugal (Hybrid)

- Developing a real-time computer vision system for automated people counting using YOLOv8-based object detection.
- Building and training deep learning models in PyTorch to detect and track individuals across video streams.
- Persisting tracking data to MongoDB and surfacing occupancy metrics through a Power BI dashboard for real-time and historical visibility.

Technologies: Python, PyTorch, YOLOv8, MongoDB, Power BI, Computer Vision

Education

BSc in Informatics Engineering
ISEP, Instituto Superior de Engenharia do Porto — Porto, Portugal

Technical Skills

Languages	Python, JavaScript, TypeScript, Java, C, C#, SQL, HTML/CSS
ML / DL	PyTorch, Scikit-Learn, Pandas, Hugging Face Transformers, PEFT, QLoRA, MLflow, YOLOv8, DeepSORT
Cloud & DevOps	AWS, Docker, Git, Linux, CI/CD
Frameworks	FastAPI, Node.js, Next.js, Spring Boot, Angular, .NET
Databases	PostgreSQL, MongoDB, Firebase, Oracle

Certifications

- AWS Certified Solutions Architect – Associate — *August 2025*
- AWS Certified Machine Learning Engineer – Associate — *June 2025*
- AWS Certified AI Practitioner – Foundational — *April 2025*
- IELTS Academic – C1 — *February 2026*

Projects

stylist. — *Image-based Outfit Compatibility Model*

- Built a siamese neural network from scratch to learn outfit compatibility in a shared embedding space, where visually compatible clothing items are mapped closer together than incompatible ones.

- Trained on the Polyvore dataset of human-curated outfits, implementing a full data preprocessing pipeline including removal of non-clothing items and mapping each item to coarse and fine-grained category labels.
- Designed embeddings as a concatenation of visual features and hierarchical category signals (coarse + fine-grained), enabling structural and semantic understanding of fashion compatibility.
- Optimised using triplet loss with hard negative mining after initial warm-up training, improving separation between compatible and incompatible outfit combinations.

Technologies: Python, PyTorch, Siamese Networks, Triplet Loss, Metric Learning, Computer Vision, Hugging Face Datasets

curate. — *Fine-tuned Outfit Recommendation Model*

- Fine-tuned Mistral-7B using QLoRA on the Polyvore dataset (21k outfit examples) for garment compatibility and outfit recommendation.
- Built full data cleaning pipeline enforcing outfit validity rules: category filtering, structural constraints, and deduplication.
- Tracked experiments with MLflow across LoRA rank and learning rate configurations; served model via FastAPI endpoint containerised with Docker on AWS ECS.

Technologies: Python, PyTorch, Hugging Face PEFT, Mistral-7B, QLoRA, MLflow, FastAPI, Docker, AWS ECS

Transformer from Scratch — *Decoder-only Music Generation Model*

- Built a decoder-only transformer from first principles in PyTorch. It has multi-head self-attention, positional encoding, residual connections, and layer norm. Trained on MIDI data for symbolic music generation.
- Designed a custom tokenisation vocabulary of 388 tokens to encode piano-roll events; built the full data pipeline from raw MIDI files using `pretty_midi`.
- Trained with cosine LR annealing, gradient clipping, and early stopping on perplexity; generation samples autoregressively with multinomial sampling over a 388-token vocabulary.

Technologies: Python, PyTorch, Transformers, NLP, pretty_midi, MIDI

headsup — *Real-Time People Counting System* *(developed during internship at DevScope)*

- Designed a real-time computer vision system for automated people counting using YOLOv8-based object detection.
- Persisted tracking data to MongoDB and surfaced occupancy metrics through a Power BI dashboard for real-time and historical visibility.

Technologies: Python, PyTorch, YOLOv8, MongoDB, Power BI, Computer Vision

LSTM Sentiment Analyser — *Review Analyser*

- Trained a 2-layer LSTM on 25,000 IMDB reviews for binary sentiment classification achieving 80% accuracy; built custom tokenisation pipeline, embedding layer, and FastAPI serving layer deployed on Hugging Face Spaces.

Technologies: Python, PyTorch, LSTM, FastAPI, Hugging Face Spaces

her. — *AI Personal Stylist*

- Built end-to-end recommendation pipeline with user style profiling, contextual preferences, and feedback analysis backed by PostgreSQL with 8+ tables.
- Integrated Hugging Face ML models for style classification via REST API with automated ETL workflows aggregating feedback data daily.

Technologies: TypeScript, Node.js, PostgreSQL, Supabase, Hugging Face, Next.js